

Question number	Answer	Mark
6(a)	The only correct answer is A <i>B is incorrect because $(120 - 74) \div 120 = 0.383333333333$</i> <i>C is incorrect because $(120 - 74) \div 120 = 0.383333333333$</i> <i>D is incorrect because $(120 - 74) \div 120 = 0.383333333333$</i>	(1)

Question number	Answer	Mark
6(b)(i)	The only correct answer is C <i>A is incorrect because $3200 \div 6.25 = 512$</i> <i>B is incorrect because $3200 \div 6.25 = 512$</i> <i>D is incorrect because $3200 \div 6.25 = 512$</i>	(1)

Question number	Answer	Additional guidance	Mark
6(b)(ii)	A description that includes three of the following points: - all the heavy metals kill (some of) the bacteria at {high concentrations / concentrations of above $200 \mu\text{g cm}^{-3}$} (1) - none of the heavy metals kill bacteria at concentrations up to $100 \mu\text{g cm}^{-3}$ (1) - different minimum concentrations are needed to kill the bacteria (1) - Cr is least {toxic / effective} / Cu is most {toxic / effective} (1) - all <i>Vibrio</i> are killed at high concentration of Cu and Ni / chromium does not kill all the <i>Vibrio</i> at {high concentrations (tested) / at concentrations of $3200 \mu\text{g cm}^{-3}$} (1)	ACCEPT mark points in context of survival ACCEPT a minimum concentration of $100 \mu\text{g cm}^{-3}$ is needed below $100 \mu\text{g cm}^{-3}$ ACCEPT <i>Vibrio</i> is most {tolerant / resistant} to Cr / <i>Vibrio</i> is least {tolerant / resistant} to Cu IGNORE efficient ACCEPT no conclusion can be made about chromium above concentrations of $3200 \mu\text{g cm}^{-3}$	(3)

Question number	Answer	Additional guidance	Mark
6(c)(i)	An explanation that includes the following points: - because if mutation occurred that made the bacteria resistant (1) - the presence of the antibiotic would have acted as a selection pressure (1) - increase in the number of bacteria with the resistance allele (1) - and the resistant bacteria would {survive / replicate} OR the non-resistant bacteria would {decrease in number / die / not replicate} (1)	ACCEPT horizontal transmission of resistance plasmid DO NOT ACCEPT {selection pressure / antibiotic} caused the mutation ACCEPT a description of the term selection pressure ACCEPT increase in resistance allele frequency	(3)

Question number	Answer	Additional guidance	Mark
6(c)(ii)	An answer that includes the following points: the bactericidal antibiotic would {kill / destroy} the bacteria AND the bacteriostatic antibiotic would prevent {growth / division} (1)	ACCEPT pathogen / microorganism for bacteria DO NOT ACCEPT virus ACCEPT prevents {protein synthesis / DNA synthesis / metabolism} keeps bacteria in the stationary phase	(1)

Question number	Answer	Additional guidance	Mark
(c)(iii)	An answer that includes the following points: - grow the bacteria without any antibiotics present and then add the antibiotics (1) + colonies will be present on bacteriostatic and not bactericidal antibiotics (1) OR - add a dye to the agar plates (1) + if the cells are stained then the antibiotic was bactericidal and if the cells are {not blue / colourless} then the antibiotic was bacteriostatic (1) OR - grow in liquid culture with the antibiotics then plate out the bacteria onto (plain) agar (1) + colonies will be present on bacteriostatic and not bactericidal antibiotics (1) OR - grow in liquid culture and then count the number of bacteria in a counting chamber using an exclusion dye (1) + if the cells are blue then the antibiotic was bactericidal and if the cells are {not blue / colourless} then the antibiotic was bacteriostatic (1) OR - add a known number of cells to a liquid culture containing antibiotics (leave and recount) (1) + if cell number stays the same then the antibiotic is bacteriostatic and if the cell number decreases it is bactericidal (1)	ACCEPT colonies will show around filter paper disc if bacteriostatic but there would be a zone of inhibition if bactericidal ACCEPT fluorescent tag / indicator + appropriate description of what would be seen	(2)

